Clinical Measurement Audits and Two-Phase Imputation: Concise Study Guide

A twelve-lesson study guide to Patrick Iben's *Statistical Inference and Two-Phase Audit Design for Contaminated Measurements in Clinical Research: The catmech R Package*, canonical manuscript paper/manuscript.Rmd (350 lines, SHA-256 8152288b1c2af2a49103543db3c9324582177b1e842365b5c5ab83cdac8c1fd0) and compiled 15-page PDF paper/manuscript.pdf (SHA-256 1d17480886d4f38c38e7cf5f900ea3c5f1ef48f3c4484fa801a72a83de05e0db).

This educational companion explains the statistical definitions, benchmark archives, and R workflows in the frozen catmech manuscript; it does not establish universal interval coverage under unmodeled misspecification, validate per-record error probabilities for individual patient adjudication on diffuse real-world data, or certify FDA 21 CFR Part 11 or ICH E9 regulatory compliance.

How to use this twelve-lesson route

This concise guide is designed for clinicians, trial statisticians, and data auditors who want the core mathematical logic, empirical findings, R function calls, and scope boundaries in twelve short lessons (S1 through S12). You can also step through these exact twelve lessons interactively in an R console by running `source('education/interactive_tutor.R'); catmech_tutor()`, or read the full thirty-unit passage-by-passage companion in COMPANION.md (PDF).

S1. The three-way clinical audit dilemma and why recorded values matter

Where we are

Abstract and Section 1, manuscript lines 1-41 (PDF pages 1-2).

What the passage is doing

When an audit re-measures a subset of records (`R_i`), trialists face three flawed conventional choices: (1) naively trusting recorded values (`W_i`) causes attenuation bias; (2) analyzing only audited rows (`gold_only`) inflates standard errors; and (3) setting unaudited `W_i` to `NA` (`reference_normal`) wastes accurate recordings. On ClinicalTrials.gov (`N = 27,208`) and NHANES (`N = 3,908`), keeping `W_i` via `catmech_reference(use_recorded = TRUE)` reduces RMSE by 43% to 48% over discarding unaudited values.

In ordinary language

Audits check a small slice of records against a trusted reference. Keeping recorded numbers cuts error nearly in half compared with throwing unchecked numbers away.

Evidence status

SCOPE and MEASURED_DATED summary across 10 frozen benchmark archives.

Do not accidentally conclude

Do not conclude that standard MAR imputation (`mice`, `Amelia`) is biased when unaudited `W_i` values are discarded; its penalty in that mode is wider confidence intervals.

Try it

Run `round(100 * (1 - 0.0409 / 0.0785), 1)` in R to verify the NHANES RMSE reduction at a 25% audit. **Answer:** `47.9`%.

S2. Two-phase contamination mixture model and marginalized corruption probabilities

Where we are

Sections 2.1-2.2, manuscript lines 42-70 (PDF pages 2-3).

What the passage is doing

Latent true values follow `T_i | X_i ~ N(X_i beta, sigma_T^2)` and recorded values follow `W_i = (1 - delta_i) T_i + delta_i (T_i + epsilon_i)` with `delta_i in {0, 1}` and `epsilon_i ~ N(mu_e, sigma_e^2)`. On unaudited rows, `catmech_contamination()` integrates out `(delta_i, T_i)` to compute the exact marginal corruption probability `p_i`. When `delta_i = 0`, `T_i = W_i` deterministically; when `delta_i = 1`, `T_i` is drawn from the precision-weighted conjugate Gaussian posterior.

In ordinary language

Each row has a hidden true value and a recorded value. The mixture model compares how well an unchecked recording fits the clean curve versus the wider error curve.

Evidence status

DEFINITION and ANALYTIC conditional posteriors for `catmech_contamination()`.

Do not accidentally conclude

Do not conclude that `reference_sd` (`sigma_ref`) is estimated from the data; it defaults to `0` (exact recounts) and is a user-declared sensitivity constant.

Try it

When `delta_i = 0` is drawn for an unaudited record in `catmech_contamination()`, what is the imputed value `T_i`? **Answer:** `T_i = W_i` deterministically.

S3. Gaussian reference imputation in C++17: use_recorded = TRUE versus FALSE

Where we are

Section 2.2, manuscript lines 71-72 (PDF pages 3-4).

What the passage is doing

`catmech_reference()` implements fast conjugate Gaussian reference imputation in C++17 (`RcppArmadillo`). With `use_recorded = TRUE` (Simple Linear MI, `cole_mi`), `W_i` enters directly as a continuous linear calibration predictor alongside `X_i` (`T_i | W_i, X_i`). With `use_recorded = FALSE` (`reference_normal`), unaudited `W_i` values are discarded.

In ordinary language

When errors are small and spread out, a fast linear regression on the audited rows calibrates the unchecked numbers without needing a mixture model.

Evidence status

DEFINITION of `catmech_reference(use_recorded = TRUE)` versus `use_recorded = FALSE`.

Do not accidentally conclude

Do not forget to pass `use_recorded = TRUE` explicitly when you want Simple Linear MI; the formal function default is `use_recorded = FALSE` for v0.3 backward compatibility.

Try it

In `Lab C16` (`education/run_companion_labs.R`), why does `use_recorded = TRUE` yield a smaller pooled standard error than `use_recorded = FALSE`? **Answer:** Because `W_i` explains substantial residual variance in `T_i` on unaudited rows.

S4. Prospective two-phase audit power planning and the NO-GO planner boundary

Where we are

Section 2.3, manuscript lines 73-84 (PDF page 4).

What the passage is doing

At audit fraction `f = n_audit / N`, the confidence interval width ratio relative to a 100% census audit is `rho(f) = sqrt(1 + k (1 - f) / f)`. `catmech_audit_power()` simulates `2,000` log-normal draws of `k_hat` to compute the power to achieve `rho <= rho_target`. In pre-specified validation (`validation/audit-plan-study`), `catmech_audit_plan()` received a `NO-GO` verdict (`11.3%-16.8%` median width error at `f = 5%`) and is documented as exploratory.

In ordinary language

The power calculator sizes an audit by simulating uncertainty in the two-phase noise factor. The empirical pilot planner missed its 10% accuracy target and stays exploratory.

Evidence status

ANALYTIC influence-function ratio and MEASURED_DATED `NO-GO` boundary on `catmech_audit_plan()`.

Do not accidentally conclude

Do not treat `catmech_audit_plan()` as a validated regulatory sample-size guarantee; its pre-specified status is `NO-GO`.

Try it

Compute `rho(0.25) = sqrt(1 + 0.27 * (1 - 0.25) / 0.25)` in R (`round(sqrt(1 + 0.27 * 3), 4)`). **Answer:** `1.3454`.

S5. Rubin combining rules and Barnard-Rubin finite-sample degrees of freedom

Where we are

Section 2.4, manuscript lines 85-94 (PDF pages 4-5).

What the passage is doing

Across `m` completions, total variance is `T = U_bar + (1 + 1/m) B` and the fraction of missing information is `lambda_hat = (1 + 1/m) diag(B) / diag(T)`. To prevent anti-conservative intervals when sample size `N`, site count `C`, or event count `d` is small, `catmech` computes Barnard-Rubin (1999) degrees of freedom `nu_BR = (nu_m * nu_obs) / (nu_m + nu_obs) < nu_com`.

In ordinary language

Old imputation formulas can claim more degrees of freedom than the study has sites or events. The Barnard-Rubin formula caps degrees of freedom below the complete-data limit.

Evidence status

ANALYTIC Rubin (1987) and Barnard-Rubin (1999) pooling formulas.

Do not accidentally conclude

Do not assume that increasing `m` to 100 lets `nu_BR` exceed `C - 1` in a clustered study; `nu_BR` is strictly bounded below `nu_obs < nu_com = C - 1`.

Try it

For `C = 10` clinical sites (`nu_com = 9`), `m = 5`, and `lambda_hat = 0.2308`, what is `nu_BR` (`Lab C10-C11`)? **Answer:** `5.3577` (strictly below `9`, whereas classical `nu_m = 75.11`).

S6. Multi-endpoint pooling across linear (CV3), GLM (CR1S), and Cox PH (Lin-Wei) models

Where we are

Section 2.4, manuscript lines 95-100 (PDF pages 5-6).

What the passage is doing

`catmech_pool_lm()` uses the delete-one-cluster jackknife (`cv3`) with `nu_com = C - 1` when clustered; `catmech_pool_glm()` uses the CR1S Huber-White-Liang-Zeger sandwich with `nu_com = C - 1`; and `catmech_pool_coxph()` uses event-based `nu_com = d - p` when unclustered or the Lin-Wei score-residual sandwich with `nu_com = C - 1` when clustered.

In ordinary language

Linear models use a leave-one-site-out jackknife variance, logistic and count models use a small-sample sandwich, and survival models count observed events.

Evidence status

ANALYTIC multi-endpoint covariance and degrees-of-freedom specifications.

Do not accidentally conclude

Do not use total sample size `N - p` for unclustered Cox proportional hazards degrees of freedom; `catmech_pool_coxph()` uses observed failure events `d - p`.

Try it

In an unclustered Cox PH model with `N = 600` patients, `d = 45` events, and `p = 3` predictors, what is `nu_com`? **Answer:** `d - p = 45 - 3 = 42`.

S7. Out-of-sample rank-PIT calibration checking and its slope-bias blind spot

Where we are

Section 2.5, manuscript lines 101-106 (PDF page 6).

What the passage is doing

`catmech_holdout_check()` runs K-fold cross-validation on audited rows and computes tie-randomized rank PIT values `u_i` against `Uniform(0, 1)`. While false alarms were `0/40` under the true model, alarm power against structural misspecification was low (`0/40` in `nonlinear_target`, `1/40` in `heavy_tail_target`) because held-out rows test observation-level prediction rather than regression slope bias.

In ordinary language

Holdout checking hides part of the audit and tests if row predictions match the hidden truth. Passing this screen does not guarantee an unbiased regression slope.

Evidence status

ANALYTIC rank-PIT formula and MEASURED_DATED diagnostic power boundary.

Do not accidentally conclude

Do not treat `miscalibration_detected = FALSE` in `catmech_holdout_check()` as proof that downstream regression slopes are unbiased.

Try it

Why does `catmech_holdout_check()` randomize exact ties (`T_{i,-k}^{(j)} = R_i^*`) using `U_i ~ Uniform(0, 1)`? **Answer:** Because the contamination mixture places a discrete point mass at `W_i` on uncorrupted rows; without tie randomization, exact matches would spike and trigger false KS alarms.

S8. Synthetic contamination benchmarks (52,950 requests) and three stress-cell failures

Where we are

Sections 3-5.1, manuscript lines 107-249 (PDF pages 6-10).

What the passage is doing

Across `52,950` synthetic requests, `catmech_contamination()` narrowed CIs by `44.3%` over complete-case analysis in `missed_flags` (`0.1842` vs `0.3309`) and prevented the `-0.2354`

linear-MI bias under gross 10x outliers (`error_gross_and_small`, `94.0%` vs `64.7%` coverage). However, it failed in three stress cells: `heavy_tail_target` (`87.3%` coverage), `nonignorable_missed` (`79.3%` coverage), and `noisy_undeclared` (`88.7%` coverage, restored to `91.8%` when `reference_sd` is declared).

In ordinary language

Under synthetic gross or unflagged errors, the mixture model stays unbiased and cuts interval width by 44%. It fails under heavy tails, non-ignorable selection, or undeclared reference noise.

Evidence status

MODELED_LOCKED synthetic benchmark results across 27 scenarios.

Do not accidentally conclude

Do not use `catmech_contamination()` when the true distribution `T_i` has heavy `t_4` tails (`heavy_tail_target`); use `catmech_reference(use_recorded = TRUE)` (`92.0%` coverage) instead.

Try it

What caused `catmech_contamination()` to under-cover (`87.3%`) in `heavy_tail_target`?
Answer: The Gaussian mixture misclassified genuine heavy-tailed extremes in `T_i` as recording errors and shrunk them toward `X_i beta`.

S9. Clustered variance (148,500 requests): CV3 jackknife and predictor-slope attenuation

Where we are

Section 5.2, manuscript lines 250-258 (PDF pages 10-11).

What the passage is doing

Across `148,500` clustered requests on simulated cohorts and 11 Rwanda HMIS indicators, `cv3` beat `cr1s` and `cr2` (which inflated widths up to `95%`). However, when the audited variable `T_i` was a **predictor** (`Y ~ T + x + stratum`), clustered random-intercept imputation (`catmech_reference(cluster = )`) attenuated slopes by up to `-16.1%` (`84.4%` coverage), whereas fitting `catmech_reference(use_recorded = TRUE)` without `cluster` and pooling with `catmech_pool_lm(cluster = )` achieved `-3.0%` bias and `95.4%` coverage.

In ordinary language

Across clinical sites, the leave-one-site-out jackknife gives the best interval coverage. However, random-intercept imputation shrinks slopes when the audited variable is a predictor.

Evidence status

MEASURED_DATED and MODELED_LOCKED clustered variance and multilevel MI evaluation.

Do not accidentally conclude

Do not pass `cluster` to `catmech_reference()` when the audited variable is a predictor in the downstream regression (`catmech_pool_lm()` warns if you do).

Try it

What is the recommended two-step R call when an audited biomarker is a predictor in a multicenter linear regression? **Answer:** Fit `catmech_reference(..., use_recorded = TRUE)` without `cluster`, then pool with `catmech_pool_lm(..., cluster = 'site')`.

S10. Public paired-discrepancy benchmarks (33,800 requests) and near-chance row error AUCs

Where we are

Section 6.1, manuscript lines 259-279 (PDF pages 11-12).

What the passage is doing

On ClinicalTrials.gov (`N = 27,208`) and NHANES 2017-2018 (`N = 3,908`), retaining recorded values reduced RMSE by `43%` to `48%` over discarding unaudited values. However, because real-world discrepancies were small relative to `T_i | X_i`, per-record corruption probabilities `p_i` had near-chance discrimination (`AUC = 0.543` on ClinicalTrials.gov and `0.500` on NHANES, versus `0.962` in synthetic `missed_flags`).

In ordinary language

On real data with diffuse errors, keeping recorded numbers cuts slope error nearly in half, but per-row error probabilities score no better than a coin flip.

Evidence status

MEASURED_DATED empirical benchmark across ClinicalTrials.gov and NHANES.

Do not accidentally conclude

Never use individual record-level corruption probabilities `p_i` from `catmech_contamination()` to adjudicate or discard specific patient records in diffuse real-world data.

Try it

What was the ROC-AUC of per-record corruption probabilities `p_i` at a 25% audit on ClinicalTrials.gov and NHANES? **Answer:** `0.543` on ClinicalTrials.gov and `0.500` on NHANES (near coin-flip chance `0.500`).

S11. Real-audit slope recovery (45,600 requests) and the methodological decision tree

Where we are

Sections 6.2-8, manuscript lines 280-328 (PDF pages 12-14).

What the passage is doing

On Rwanda HMIS facility recounts (`45,600` requests), Rule 2 comparing `catmech_contamination()` against `Linear MI` was `INCONCLUSIVE`: the mixture had lower point RMSE on `PNC` (`0.053` vs `0.072`) and `Iron` (`0.112` vs `0.156`), but failed 95% coverage on all three primary slopes (`71.6%` on `PNC`, `66.6%` on `Syphilis`, `35.4%` on `Iron`) and failed the `>= 90%` return rate on two. Therefore, Section 7 recommends `catmech_contamination()` only for gross, separable outliers and `catmech_reference(use_recorded = TRUE)` for moderate or diffuse real-world discrepancies.

In ordinary language

On Rwanda health-center recounts, the mixture model had lower point error on two slopes but failed interval coverage. For everyday moderate errors, use simple linear imputation.

Evidence status

MEASURED_DATED real-audit evaluation (`INCONCLUSIVE` Rule 2) and INTERPRETIVE decision tree.

Do not accidentally conclude

Do not claim that `catmech_contamination()` outperformed Simple Linear MI on real-world audit intervals; its interval advantage remains synthetic-only.

Try it

Under the Section 7 decision tree, what two checks must pass before trusting `catmech_contamination()` on a dataset with suspected gross errors? **Answer:** A `>= 90%` interval return rate and positive held-out Brier/RMSE skill over raw recorded values in `catmech_holdout_check()`.

S12. Provenance reporting (catmech_audit_trail), software QA, and regulatory boundaries

Where we are

Sections 9-12, manuscript lines 329-350 (PDF pages 14-15).

What the passage is doing

`catmech_audit_trail()` outputs a provenance record (Markdown, JSON, or HTML) recording R/OS metadata, SHA-256 hashes of the input data, assumption specification, and all `m` completed datasets, MCMC convergence status, and pooled tables. This supports documentation under FDA 21 CFR Part 11 and ICH E9 without certifying compliance. The package is MIT-licensed and verified by `1,754` unit tests (`0` failures/warnings/skips), `R CMD check --as-cran` (`0` errors/warnings), and `500,000+` replicate checks.

In ordinary language

The audit trail function writes a report with SHA-256 data hashes, model settings, and convergence status. It supports documentation but does not certify legal compliance.

Evidence status

DEFINITION, SCOPE, and MEASURED_DATED software verification summary.

Do not accidentally conclude

Do not represent `catmech_audit_trail()` as a legal compliance certification; regulatory compliance depends on the surrounding validated system, access controls, and procedures.

Try it

Run `Lab C28` in `education/run_companion_labs.R`. What default title and footer heading does `catmech_audit_trail()` print? **Answer:** Default title `Catmech Audit Trail & Provenance Report` and footer heading `### Audit Trail & Provenance Record`.